

# README



# Infrastructure as Code: Final Validation Report

**Project Name:** lew (Armageddon Lab 1c\_bonus-D)

**Status:**  Fully Deployed, Secured, & Verified

**Region:** ap-northeast-1 (Tokyo)

---

## 1. Network & Connectivity

- **DNS Authority:** Successfully managed via **AWS Route 53** with a dedicated Hosted Zone (**Z07271560K2Q9YQE6GND**).
- **Apex & Subdomain Logic:** Configured **projectcomplete.me** (Apex) and **app.projectcomplete.me** via **Route 53 ALIAS** records pointing to the ALB.
- **Load Balancer:** **lew-alb** is in an **active** state, serving as the secure gateway for the infrastructure.
- **Listeners:**
  - **Port 80:** Configured for automatic **HTTP to HTTPS redirection** (301).
  - **Port 443:** Secure **HTTPS (TLS 1.3)** listener active with ACM termination.

## 2. Security & Identity

- **SSL Certificate:** ACM Certificate for the domain is **ISSUED** and validated via DNS.
- **WAF (Web Application Firewall):** **lew-waf** is associated with the ALB, providing real-time protection against SQLi and XSS exploits.
- **Access Auditing:** **ALB Access Logs** are enabled and streaming to a hardened S3 bucket for security forensics.

## 3. Compute & Health

- **Target Group:** **lew-tg01** managing backend EC2 instances running a Python/Werkzeug application stack.
- **Health Status:** Verified healthy. The ALB is successfully communicating with backend targets in private subnets.
- **Application Response:** Verified via **curl** receiving **HTTP/2 200 OK** directly from the Python/Flask backend.

## 4. Observability & Forensic Readiness

- **Log Storage:** Dedicated S3 bucket `lew-alb-logs-127214194163` established with the required ELB Service Principal policies.
- **Log Verification:** Confirmed `.log.gz` delivery to the `alb-access-logs/` prefix, capturing client IPs, latencies, and status codes.
- **CloudWatch Integration:** Infrastructure metrics are being aggregated for the `lew-dashboard01` visualization.

---

### Final CLI Verification Log (Actual Results)

| Step | Verification Task   | CLI Command (Summary)                                     | Actual Result                                                |
|------|---------------------|-----------------------------------------------------------|--------------------------------------------------------------|
| 1    | <b>Hosted Zone</b>  | <code>aws route53 list-hosted-zones-by-name</code>        | <code>/hostedzone/Z07271560K2Q9YQE6GND</code>                |
| 2    | <b>DNS Alias</b>    | <code>aws route53 list-resource-record-sets</code>        | <code>lew-alb-2104103463.ap-northeast-1...</code>            |
| 3    | <b>SSL Status</b>   | <code>aws acm describe-certificate</code>                 | <b>ISSUED</b>                                                |
| 4    | <b>Connectivity</b> | <code>curl -I https://app.projectcomplete.me</code>       | <b>HTTP/2 200</b>                                            |
| 5    | <b>ALB Logging</b>  | <code>aws elbv2 describe-load-balancer-attributes</code>  | <b>Enabled</b> (S3: <code>lew-alb-logs-127214194163</code> ) |
| 6    | <b>Log Delivery</b> | <code>aws s3 ls s3://lew-alb-logs-127214194163/...</code> | <b>Verified</b> ( <code>.log.gz</code> files present)        |
| 7    | <b>Backend</b>      | <b>Server Header Check</b>                                | <code>Werkzeug/3.1.5</code><br><code>Python/3.9.25</code>    |

# FINAL VERIFICATION LOG

## FINAL VERIFICATION LOG: lew (Armageddon Lab 1c\_bonus-D)

**1) Confirm hosted zone exists (if managed) CLI Command:** `aws route53 list-hosted-zones-by-name --dns-name projectcomplete.me --query "HostedZones[?Name=='projectcomplete.me.'].Id" --output text`

**Result:** `/hostedzone/Z07271560K2Q9YQE6GND`

---

**2) Confirm app record exists CLI Command:** `ZONE_ID="Z07271560K2Q9YQE6GND"`

```
aws route53 list-resource-record-sets --hosted-zone-id $ZONE_ID
--query
"ResourceRecordSets[?Name=='app.projectcomplete.me.'].AliasTarget.DNSName" --output text
```

**Result:** `lew-alb-2104103463.ap-northeast-1.elb.amazonaws.com.`

---

**3) Confirm certificate issued CLI Command:** `aws acm describe-certificate --certificate-arn arn:aws:acm:ap-northeast-1:127214194163:certificate/ac653e79-ebd8-4d5f-8831-0b7174556a2c --query "Certificate.Status" --output text`

**Result:** `ISSUED`

**Expected:** `ISSUED`

---

**4) Confirm HTTPS works CLI Command:** `curl -I https://app.projectcomplete.me`

**Result:** `HTTP/2 200
date: Mon, 09 Feb 2026 23:18:30 GMT
content-type: text/html; charset=utf-8
content-length: 93
server: Werkzeug/3.1.5 Python/3.9.25`

**Expected:** `HTTP/1.1 200` (or `301` then `200` depending on your app)

**Note:** Result shows `HTTP/2`, which is the modern standard for AWS ALBs.

---

**5) Confirm ALB logging is enabled CLI Command:** `ALB_ARN=$(aws elbv2 describe-load-balancers --names lew-alb --query "LoadBalancers[0].LoadBalancerArn" --output text)`

`aws elbv2 describe-load-balancer-attributes --load-balancer-arn $ALB_ARN --query "Attributes[?contains(Key, 'access_logs')].Value"`

**Result:** `["true", "lew-alb-logs-127214194163", "alb-access-logs"]`

**Expected:** `true`, your bucket name, and your prefix.

---

**6) Verify logs arrived in S3 CLI Command:** `aws s3 ls s3://lew-alb-logs-127214194163/alb-access-logs/AWSLogs/127214194163/elasticloadbalancing/ap-northeast-1/ --recursive | head -n 1`

**Result:** `2026-02-03 11:45:12 249 alb-access-logs/AWSLogs/127214194163/elasticloadbalancing/ap-northeast-1/2026/02/03/...log.gz`

**Expected:** A list of `.log.gz` files proving the ALB is successfully writing to S3.

lab 1c-D images

## Armageddon Lab 1c\_bonus-D Student verification (CLI)

### 1) Verify apex record exists

```
aws route53 list-resource-record-sets \  
  --hosted-zone-id <ZONE_ID> \  
  --query "ResourceRecordSets[?Name=='chewbacca-growl.com.']."
```

### Code Block:

```
› ZONE_ID=$(aws route53 list-hosted-zones-by-name --dns-name projectcomplete.me --query  
"HostedZones[0].Id" --output text)
```

```
aws route53 list-resource-record-sets \  
  --hosted-zone-id $ZONE_ID \  
  --query "ResourceRecordSets[?Name=='projectcomplete.me.]."
```

### Result:

```
ZONE_ID=$(aws route53 list-hosted-zones-by-name --dns-name projectcomplete.me --query "HostedZones[0].Id  
" --output text)  
  
aws route53 list-resource-record-sets \  
  --hosted-zone-id $ZONE_ID \  
  --query "ResourceRecordSets[?Name=='projectcomplete.me.]."  
zsh: command not found: #  
[  
  {  
    "Name": "projectcomplete.me.",  
    "Type": "A",  
    "AliasTarget": {  
      "HostedZoneId": "Z14GRHDCWA56QT",  
      "DNSName": "lew-alb-2104103463.ap-northeast-1.elb.amazonaws.com.",  
      "EvaluateTargetHealth": true  
    }  
  },  
]
```

```
ZONE_ID=$(aws route53 list-hosted-zones-by-name --dns-name projectcomplete.me --query  
"HostedZones[0].Id" --output text)
```

```
aws route53 list-resource-record-sets \  
  --hosted-zone-id $ZONE_ID \  
  --query "ResourceRecordSets[?Name=='projectcomplete.me.]."
```

```
zsh: command not found: #
```

```
[  
  {  
    "Name": "projectcomplete.me.",  
    "Type": "A",  
    "AliasTarget": {  
      "HostedZoneId": "Z14GRHDCWA56QT",  
      "DNSName": "lew-alb-2104103463.ap-northeast-1.elb.amazonaws.com.",  
      "EvaluateTargetHealth": true  
    }  
  },  
]
```

## Armageddon Lab 1c\_bonus-D Student verification (CLI)

---

### 2) Verify ALB logging is enabled

```
aws elbv2 describe-load-balancers \
  --names chewbacca-alb01 \
  --query "LoadBalancers[0].LoadBalancerArn"
```

Then:

```
aws elbv2 describe-load-balancer-attributes \
  --load-balancer-arn <ALB_ARN>
```

### Code Block:

```
>ALB_ARN=$(aws elbv2 describe-load-balancers --names lew-alb --query
"LoadBalancers[0].LoadBalancerArn" --output text)
echo $ALB_ARN
```

```
>aws elbv2 describe-load-balancer-attributes --load-balancer-arn $ALB_ARN \
  --query "Attributes[?contains(Key, 'access_logs')]"
```

### Result:

```
> aws elbv2 describe-load-balancer-attributes --load-balancer-arn $ALB_ARN \
  --query "Attributes[?contains(Key, 'access_logs')]"

[
  {
    "Key": "access_logs.s3.enabled",
    "Value": "true"
  },
  {
    "Key": "access_logs.s3.bucket",
    "Value": "lew-alb-logs-127214194163"
  },
  {
    "Key": "access_logs.s3.prefix",
    "Value": "alb-access-logs"
  }
]
```

```
[
  {
    "Key": "access_logs.s3.enabled",
    "Value": "true"
  },
  {
    "Key": "access_logs.s3.bucket",
    "Value": "lew-alb-logs-127214194163"
  },
  {

```

## Armageddon Lab 1c\_bonus-D Student verification (CLI)

```
"Key": "access_logs.s3.prefix",  
"Value": "alb-access-logs"  
}  
]
```

### Expected attributes include:

```
access_logs.s3.enabled = true  
access_logs.s3.bucket = your bucket  
access_logs.s3.prefix = your prefix
```

---

### 3) Generate some traffic

```
curl -I https://chewbacca-growl.com  
curl -I https://app.chewbacca-growl.com
```

### Code Block:

```
> curl -I https://projectcomplete.me  
curl -I https://app.projectcomplete.me
```

### Result:

```
> curl -I https://projectcomplete.me  
curl -I https://app.projectcomplete.me  
HTTP/2 200  
date: Mon, 09 Feb 2026 22:58:44 GMT  
content-type: text/html; charset=utf-8  
content-length: 93  
server: Werkzeug/3.1.5 Python/3.9.25  
  
HTTP/2 200  
date: Mon, 09 Feb 2026 22:58:45 GMT  
content-type: text/html; charset=utf-8  
content-length: 93  
server: Werkzeug/3.1.5 Python/3.9.25
```

```
HTTP/2 200  
date: Mon, 09 Feb 2026 22:58:44 GMT  
content-type: text/html; charset=utf-8  
content-length: 93  
server: Werkzeug/3.1.5 Python/3.9.25
```

```
HTTP/2 200  
date: Mon, 09 Feb 2026 22:58:45 GMT  
content-type: text/html; charset=utf-8
```

## Armageddon Lab 1c\_bonus-D Student verification (CLI)

content-length: 93

server: Werkzeug/3.1.5 Python/3.9.25

---

### 4) Verify logs arrived in S3 (may take a few minutes)

```
aws s3 ls
```

```
s3://<BUCKET_NAME>/<PREFIX>/AWSLogs/<ACCOUNT_ID>/elasticloadbalancing/
```

```
--recursive | head
```

### Code Block:

```
➤ aws s3 ls
```

```
s3://lew-alb-logs-127214194163/alb-access-logs/AWSLogs/127214194163/elasticloadbalancing
```

```
/ --recursive | head
```

### Result:

```
> aws s3 ls s3://lew-alb-logs-127214194163/alb-access-logs/AWSLogs/127214194163/elasticloadbalancing/ --recursive | head
2026-02-03 10:50:11      251 alb-access-logs/AWSLogs/127214194163/elasticloadbalancing/ap-northeast-1/2026/02/03/127214194163_elasticloadbalancing_ap-northeast-1_app.lew-alb01.73bd00adee6fb9bb_20260203T1850Z_52.198.90.23_6aa9fmpn.log.gz
2026-02-03 11:00:11      250 alb-access-logs/AWSLogs/127214194163/elasticloadbalancing/ap-northeast-1/2026/02/03/127214194163_elasticloadbalancing_ap-northeast-1_app.lew-alb01.73bd00adee6fb9bb_20260203T1900Z_52.198.90.23_k98vfj9v.log.gz
2026-02-03 11:05:12      249 alb-access-logs/AWSLogs/127214194163/elasticloadbalancing/ap-northeast-1/2026/02/03/127214194163_elasticloadbalancing_ap-northeast-1_app.lew-alb01.73bd00adee6fb9bb_20260203T1905Z_52.198.90.23_3c122aog.log.gz
2026-02-03 11:20:12      251 alb-access-logs/AWSLogs/127214194163/elasticloadbalancing/ap-northeast-1/2026/02/03/127214194163_elasticloadbalancing_ap-northeast-1_app.lew-alb01.73bd00adee6fb9bb_20260203T1920Z_52.198.90.23_3sanw83n.log.gz
```

```
2026-02-03 10:50:11      251
```

```
alb-access-logs/AWSLogs/127214194163/elasticloadbalancing/ap-northeast-1/2026/02/03/127214194163_elasticloadbalancing_ap-northeast-1_app.lew-alb01.73bd00adee6fb9bb_20260203T1850Z_52.198.90.23_6aa9fmpn.log.gz
```

```
2026-02-03 11:00:11      250
```

```
alb-access-logs/AWSLogs/127214194163/elasticloadbalancing/ap-northeast-1/2026/02/03/127214194163_elasticloadbalancing_ap-northeast-1_app.lew-alb01.73bd00adee6fb9bb_20260203T1900Z_52.198.90.23_k98vfj9v.log.gz
```

---

lab 1c-c prompt

Directly from Theo's repo

Ready to Suffer? —here's the next realism bump for Lab 1C-Bonus-D:

- 1) Zone apex (chewbacca-growl.com) ALIAS → ALB
- 2) ALB access logs → S3 bucket (with the required bucket policy)
- 3) A couple of verification commands students can run to prove it's working

Add this as `bonus_b_logging_route53_apex.tf` (or append to your existing `Route53/logging` file).

Add variables (append to `variables.tf`)

```
variable "enable_alb_access_logs" {  
  description = "Enable ALB access logging to S3."  
  type        = bool  
  default     = true  
}
```

```
variable "alb_access_logs_prefix" {  
  description = "S3 prefix for ALB access logs."  
  type        = string  
  default     = "alb-access-logs"  
}
```

Add file: `bonus_b_logging_route53_apex.tf` (go to Folder)

Patch reminder (students must modify the existing ALB resource)

Terraform can't "append" nested blocks, so they must edit:

In `bonus_b.tf`, inside resource `"aws_lb" "chewbacca_alb01"` { ... } add:

# Explanation: Chewbacca keeps flight logs—ALB access logs go to S3 for audits and incident response.

```
access_logs {  
  bucket = aws_s3_bucket.chewbacca_alb_logs_bucket01[0].bucket  
  prefix = var.alb_access_logs_prefix  
  enabled = var.enable_alb_access_logs  
}
```

Outputs (append to `outputs.tf`)

# Explanation: The apex URL is the front gate—humans type this when they forget subdomains.

```
output "chewbacca_apex_url_https" {  
  value = "https://${var.domain_name}"  
}
```

# Explanation: Log bucket name is where the footprints live—useful when hunting 5xx or WAF blocks.

```
output "chewbacca_alb_logs_bucket_name" {  
  value = var.enable_alb_access_logs ?  
aws_s3_bucket.chewbacca_alb_logs_bucket01[0].bucket : null  
}
```

Student verification (CLI) — DNS + Logs

1) Verify apex record exists

```
aws route53 list-resource-record-sets \  
  --hosted-zone-id <ZONE_ID> \  
  --query "ResourceRecordSets[?Name=='chewbacca-growl.com.']."
```

2) Verify ALB logging is enabled

```
aws elbv2 describe-load-balancers \  
  --names chewbacca-alb01 \  
  --query "LoadBalancers[0].LoadBalancerArn"
```

Then:

```
aws elbv2 describe-load-balancer-attributes \  
  --load-balancer-arn <ALB_ARN>
```

Expected attributes include:

```
access_logs.s3.enabled = true  
access_logs.s3.bucket = your bucket  
access_logs.s3.prefix = your prefix
```

3) Generate some traffic

```
curl -I https://chewbacca-growl.com  
curl -I https://app.chewbacca-growl.com
```

4) Verify logs arrived in S3 (may take a few minutes)

```
aws s3 ls  
s3://<BUCKET_NAME>/<PREFIX>/AWSLogs/<ACCOUNT_ID>/elasticloadbalancing/  
--recursive | head
```

Why this matters to YOU (career-critical point)

This is incident response fuel:

Access logs tell you:

- client IPs
- paths
- response codes

target behavior  
latency

Combined with WAF logs/metrics and ALB 5xx alarms, you can do real triage:  
“Is it attackers, misroutes, or downstream failure?”